Winston Zhang

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EDUCATION

The University of Sydney — Expected Graduation: 2025
Bachelor of Mechanical Engineering Honours (Fluid Mechanics) — *2021 - Current*

EXPERIENCE

Sydney Motorsports Formula Student Team — Sydney, Australia
CAD | Aerodynamic Design | CFD Validation | Composite Manufacturing
Aerodynamics Engineer — *July 2022 - December 2023*
Aerodynamics Section Lead — *December 2023 - February 2025*

Dovetail Electric Aviation — Sydney, Australia
CAD | CFD Validation | Reverse Engineering
Computational Fluid Dynamics Intern — *September 2023 - January 2024*

TECHNICAL SKILLS

Composite Manufacturing - Prepreg and Hand Layup

- **Hand Layup** - Fabricated the aerodynamic package for Sydney Motorsports including the nosecone, front wing, side-pods and rear wing.
- **Mold Manufacturing** - Manufactured the mold for the front wing and side-pods with the use of a laser cutter and a 3D printer.

Computer Aided Design - SOLIDWORKS and Autodesk Inventor

- **Parametric Design** - CAD for both Sydney Motorsports and Dovetail are fully parametric and dimensioned to be easily modified without breaking the larger assembly. Parts designed include:
  - **FSAE front wing** - General design as well as the rib and spar structure
  - **Nacelle and propeller** - Re-made the nacelle, propeller and cooling duct geometry from 3D scanned files for CFD simulations.
  - **Gearbox and general structure for a robot arm** - Designed a 6 axis robot arm from scratch.

Numerical Simulation - SOLIDWORKS and ANSYS

- **Semi-automated CFD Workflow** - Part of the development of the CFD workflow for Sydney Motorsports to enable streamlined sensitivity studies.
- **Topology Study** - Created optimized designs for the swannecks for Sydney Motorsport's wings as well as their internal rib strucutres.
- **Cooling Duct Optimization | Propeller Pitch Validation** - Optimized the cooling duct for Dovetail to minimize drag at design conditions. Validated the 3D scanned geometry of Dovetail's propeller geometry against the spec sheet.
- **Thermal Comfort** - Simulated a restaurant environment to assist with HVAD design to ensure the thermal comfort of the inhabitants.

Programming - MATLAB and Python

- **Matlab** - Blade Element Analysis, custom FEM and CFD code
- **Python** - Custom CFD code, Robot Operating System (ROS2)

TECHNICAL PROJECTS

- **Homelab Server** - Acts as a media server and custom cloud storage solution.
- **Robotic Arm** - 6 axis robotic arm with NEMA motors, belt drive and cycloidal gearbox.
- **Custom PC Peripherals** - Designed a 32 gram fingertip mouse and put together a lily58 split ortholinear keyboard and designed a case.
- **UNIX** - Daily drove dual boot windows/linux systems with various distros such as Arch and various Ubuntu variations.